

Particulate Matter Monitoring & Health Effects

Issue 2026-08-24 | Entry window 24 Aug 2026 | 9 in scope, 21 screened out

9

RECORDS IN SCOPE
(21 SCREENED OUT)

4

OF 10 CLUSTERS
OCCUPIED

2

ABSTRACT-FREE
DEPOSITS (22%)

3

ATMOS. CHEM. PHYS.
RECORDS (A FIRST)

2

INTERVAL-BEARING
EXPOSURE ESTIMATES

Signal of the day

The Crossref by-ISSN leg finally caught *Atmospheric Chemistry and Physics*, and the first four records it delivered include the cleanest quantification this watch has seen of the error a light-scattering sensor makes when it cannot tell a swollen aerosol particle from an activated droplet.

Neuberger et al. (*Atmos Chem Phys*) analysed the FAIRARI 2021/2022 fog campaign at San Pietro Capofiume in the Po Valley with κ -Köhler theory and the MIMICA large-eddy simulation. **Failing to account for hydrated, non-activated particles inflated the inferred effective diameter by 81 % (11.6 to 21.0 μm) and cut the inferred cloud droplet number concentration by 87 % (97.4 to 12.4 cm^{-3}), while hydrated particles supplied 21 % of liquid water content and accounted for 36 % of sub-kilometre visibility events with no droplet activation at all.** Median κ separated cleanly by population — fog residuals 0.45, interstitial 0.40, out-of-fog aerosol 0.34. This is a fog-microphysics paper, but the mechanism is exactly the one that governs low-cost optical PM sensors at high relative humidity: the instrument sizes a water-coated particle and reports the mass of a dry one. The literature's usual answer is an empirical RH correction fitted to a co-location period, which implicitly assumes a fixed κ ; Neuberger et al. measure κ varying by a third *within a single site and campaign* depending on which population the particle belongs to. A humidity correction estimated during one aerosol regime should therefore not be expected to transfer to another, and the residual after correction is not noise — it is composition.

What changed today

- **A structural coverage gap closed, at least for one day.** The sweep caught **four *Atmospheric Chemistry and Physics* deposits** — the first ACP records it has ever returned — of which **three shipped**; the fourth, a Jungfrauoch radiocarbon CH_4/CO_2 study, is gas-phase only and was screened out. ACP is one of the two journals the source audit named as structurally invisible to PubMed, so until now the watch has been blind to it except through Consensus, which is relevance-ranked rather than date-windowed and therefore resurfaces old work. **This is a harvest-architecture result, not a literature result**, and it should be read as such: the by-ISSN Crossref leg is now demonstrably delivering the instrumentation-side coverage it was added for. Whether it does so reliably needs several more days before it can be claimed.
- **Consensus returned six on-topic sensor and exposure-modelling papers and every one of them was a duplicate.** Low-cost sensor calibration, recalibration cadence, calibration-strategy review, satellite-surface fusion in North-East India, an urban-rural balanced estimation framework for China, and a WRF-Chem/satellite/ground fusion scheme — all six were already in *seen*. *json*, shipped between 26 July and 5 August. The dedupe key caught all six before a single word was written about them. This is the expected behaviour of a relevance-ranked source against a date-windowed pipeline, and it is worth stating plainly that on a thin day Consensus adds nothing to recall; its value is in the topics PubMed cannot index, not in the recency of what it returns.
- **Two mechanistic lung-cancer records from the same Chinese journal issue, pointing at the same exposure from opposite ends.** Li & Li review GPER as the receptor through which $\text{PM}_{2.5}$'s estrogen-like organics and metal ions drive MAPK/ERK and PI3K/AKT signalling; the Wumeng Mountain synthesis assembles the exposure-component-effect-molecular chain for household smoky-coal lung cancer in the Yunnan-Guizhou Plateau, where the disease concentrates in **never-smoking women** with a **distinctive driver-gene spectrum** (low classical EGFR 19del/L858R, elevated G719X, S768I, KRAS G12C, ALK and ROS1 fusions). Both are narrative reviews with no pooled estimate and no formal search protocol reported in the deposited abstract, so neither carries evidential weight on its own. What they carry jointly is a testable claim: that a specific source-resolved PM composition maps to a specific somatic mutation spectrum. That is the kind of claim source-apportionment-capable distributed sensing could eventually address and mass-only networks cannot.
- **The only intervals in the window are on self-reported exposure categories.** Aulia Anwar et al.'s North Sumatra case-control (240/240) reports **OR 2.53 (95 % CI 1.45–4.40)** for an unhealthy indoor environment and **OR 1.91 (1.13–3.25)** for occupational exposure. Neither is a measured particulate concentration — both are questionnaire categories — so the forest plot names the exposure metric on its own axis rather than implying a PM contrast. The same model's age ORs are demographic covariates and are excluded from the figure for the reason

recorded on 20 and 23 August.

- **Two abstract-free Elsevier deposits again, both from *Atmospheric Environment*.** A machine-learning wind-correction comparison for coupled WRF–CMAQ (dynamic assimilation vs static replacement) and a particulate PAH dry-deposition study in the Bohai Sea Strait. Both ship as *Metadata only*. Scholar Gateway was queried for the wind-correction paper and cannot help — its corpus ends May 2026 — and Consensus has not indexed either. The Elsevier abstract backlog first noted on 8 August is therefore still open and now spans three weeks.

Corpus at a glance

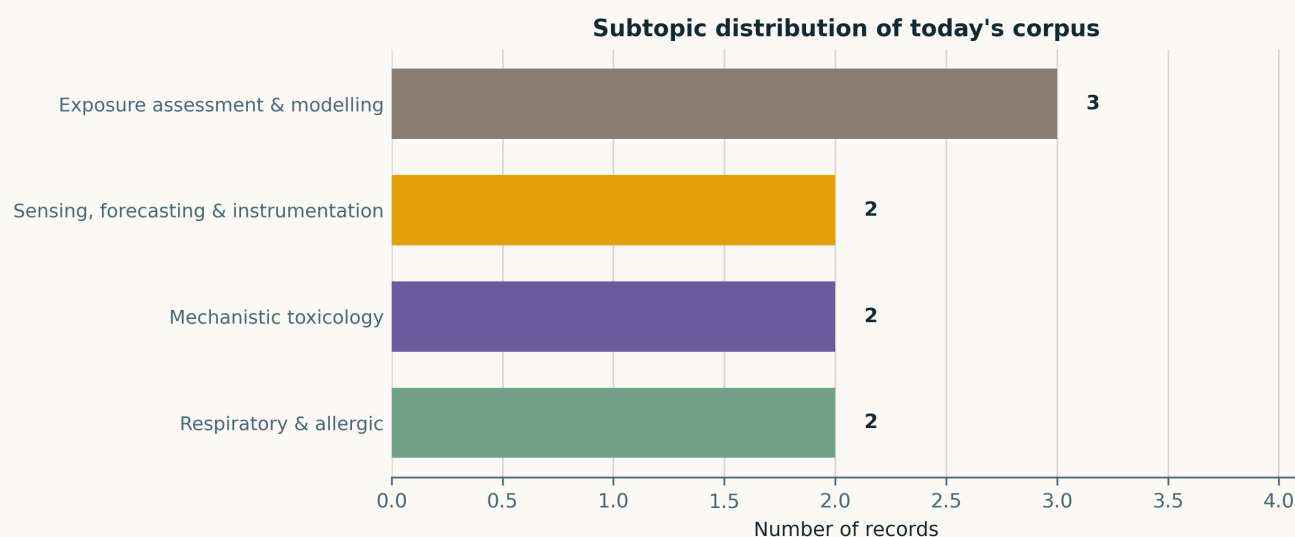


Fig. Records per subtopic cluster. **Four of ten clusters are occupied** — exposure assessment & modelling 3, sensing/instrumentation 2, mechanistic toxicology 2, respiratory & allergic 2. The six empty clusters are cardiovascular & metabolic, neuro / mental health, reproductive & developmental, occupational & indoor, burden / policy & mitigation, and other clinical endpoints. Yesterday's neuro pair and burden record have no counterpart today; the balance has shifted from clinical epidemiology toward atmospheric process and instrumentation, which is what the ACP arrivals do to a nine-record day.

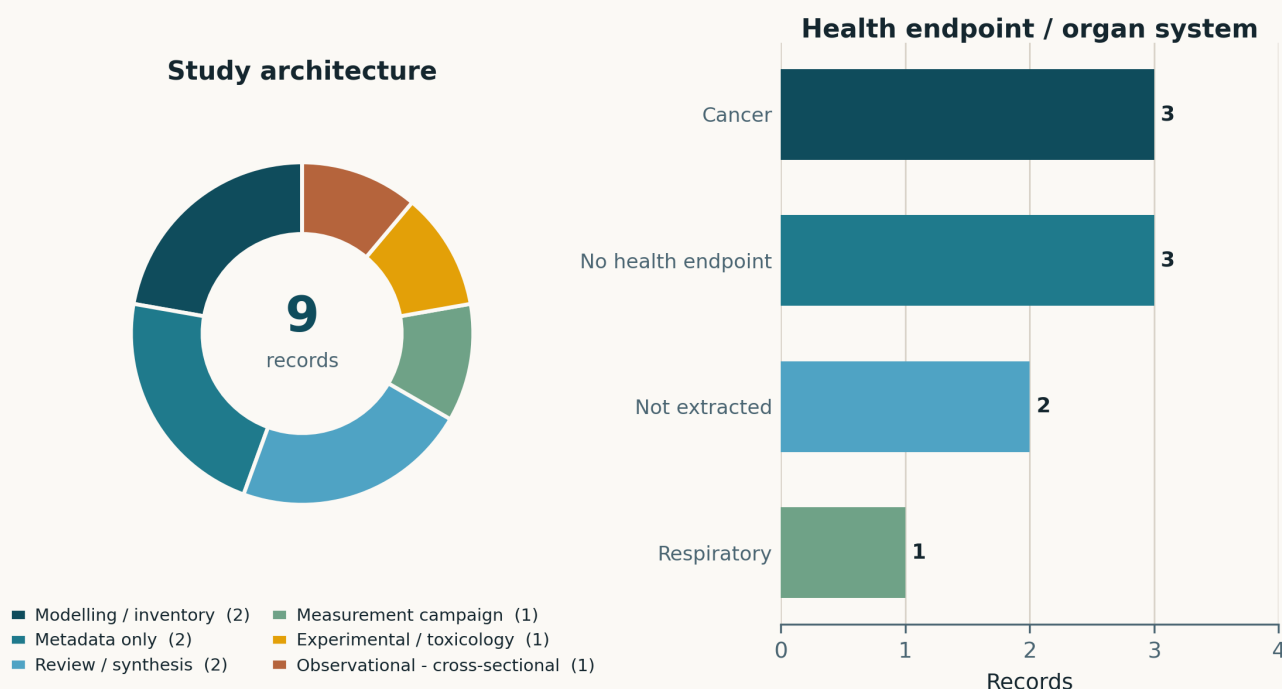


Fig. Left — study architecture, grouped. *Modelling / inventory* 2, *Metadata only* 2, *Review / synthesis* 2, and one each of measurement campaign, experimental / toxicology and observational cross-sectional — six architectures over nine records, so the panel is close to flat by construction. Right — health endpoint. **Cancer** 3 (both lung-cancer reviews plus the North Sumatra case-control), respiratory 1, and **five of nine records carry no health endpoint at all**: two instrument/atmospheric-process papers, one exposure-modelling paper, and the two metadata-only deposits whose endpoint the publisher did not deposit. Those five are split across two bars — *No health endpoint* 3 and *Not extracted* 2 — rather than pooled, so the panel understates the no-endpoint share on a first reading. Separately, the triage tiers are one A (Neuberger et al.), four B and four C; the C block is the two abstract-free deposits plus the two records whose exposure is a questionnaire category or an undeposited aerosol dose.

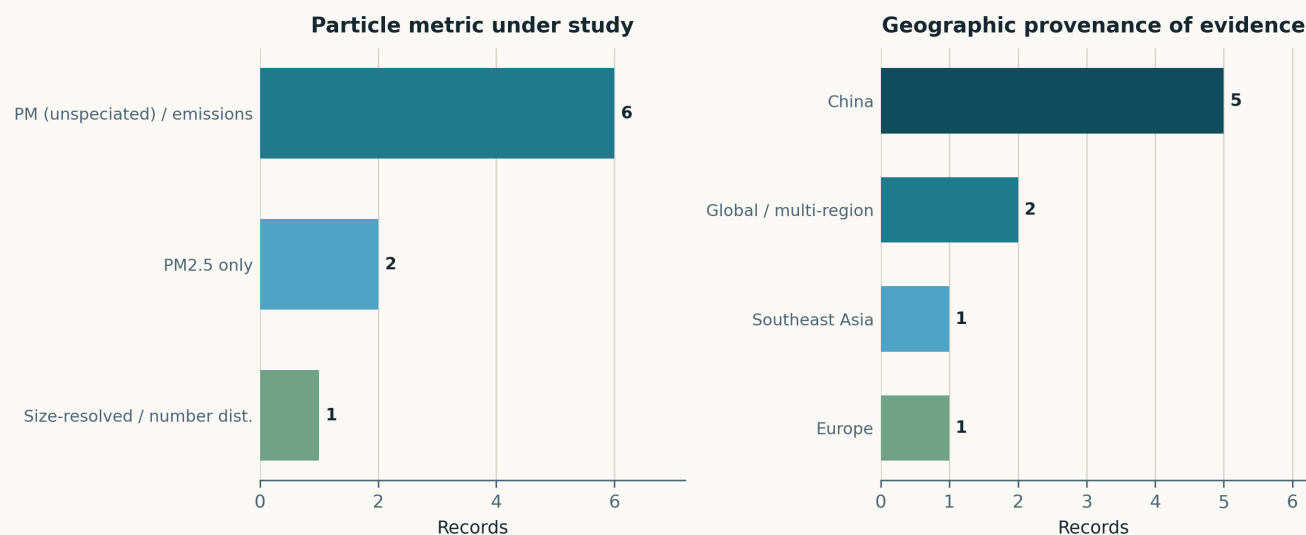


Fig. Left — particle metric, as binned by the coarse grouper: PM (*unspeciated*) / emissions 6, PM_{2.5} only 2, size-resolved / number distribution 1. **The two records in the PM_{2.5} bar do not measure PM_{2.5} mass** — they are binned there because their metric strings contain the token: Li & Li treat PM_{2.5}-borne estrogenic organics and metal ions as receptor ligands, and the Wumeng synthesis names smoky-coal PM_{2.5}/black carbon inside a PAH-dominated mixture. Read literally, **no record this window uses gravimetric PM_{2.5} as its exposure variable**; the metrics actually in play are hygroscopicity parameter and hydrated diameter, nanoparticle growth rate, isoprene-SOA mass and subspecies composition, particle-bound PAH, source-resolved coal-smoke composition, an inhaled AlCl₃ aerosol and a self-reported household-environment category. Right — geographic provenance. **China 5**, Europe 1 (Po Valley), Southeast Asia 1 (North Sumatra), multi-region 2 — and the multi-region bar is one genuinely pooled multi-continent growth-rate dataset plus one metadata-only record whose study region the publisher did not deposit, not two global studies.

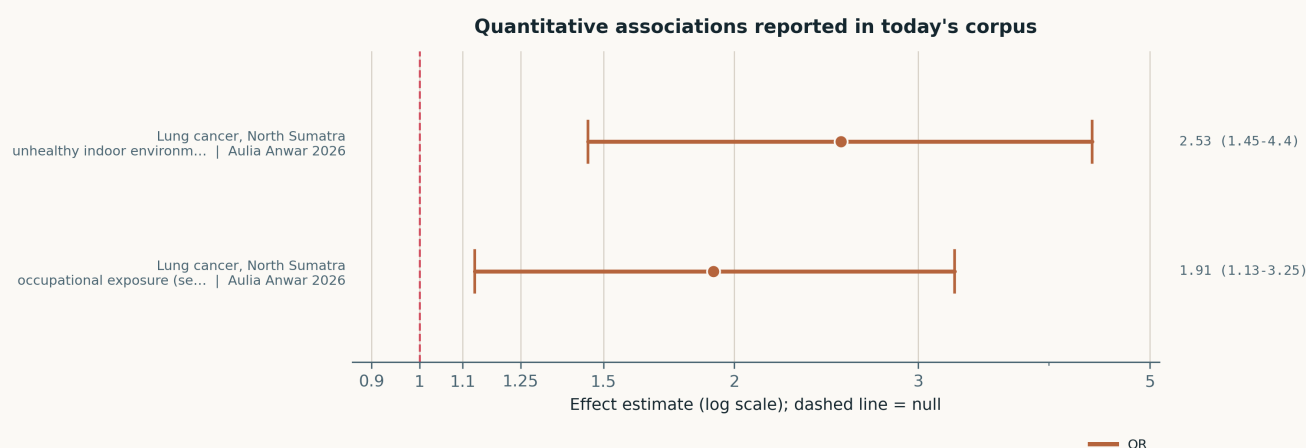


Fig. The window's only interval-bearing exposure estimates, both from the same multivariate model in Aulia Anwar et al. (240 cases / 240 controls, North Sumatra). **Both exposures are self-reported questionnaire categories, not measured concentrations**, and the axis is labelled accordingly. Two points from one study is not a synthesis and no pooled estimate is implied or drawn.

Life-course windows addressed today (records may span several stages)

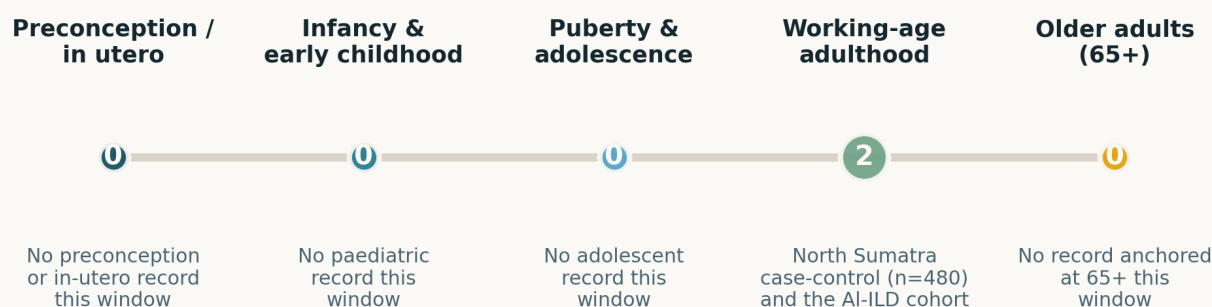


Fig. Life-course windows addressed, counted only where a record's study population is explicitly anchored at that stage. **One of five windows occupied**, and it is working-age adulthood: the North Sumatra case-control and the aluminium-ILD supportive cohort. The perinatal, paediatric and adolescent windows are empty for the second consecutive issue, and unlike yesterday the 65+ window is empty too — no record this window studies an older population.

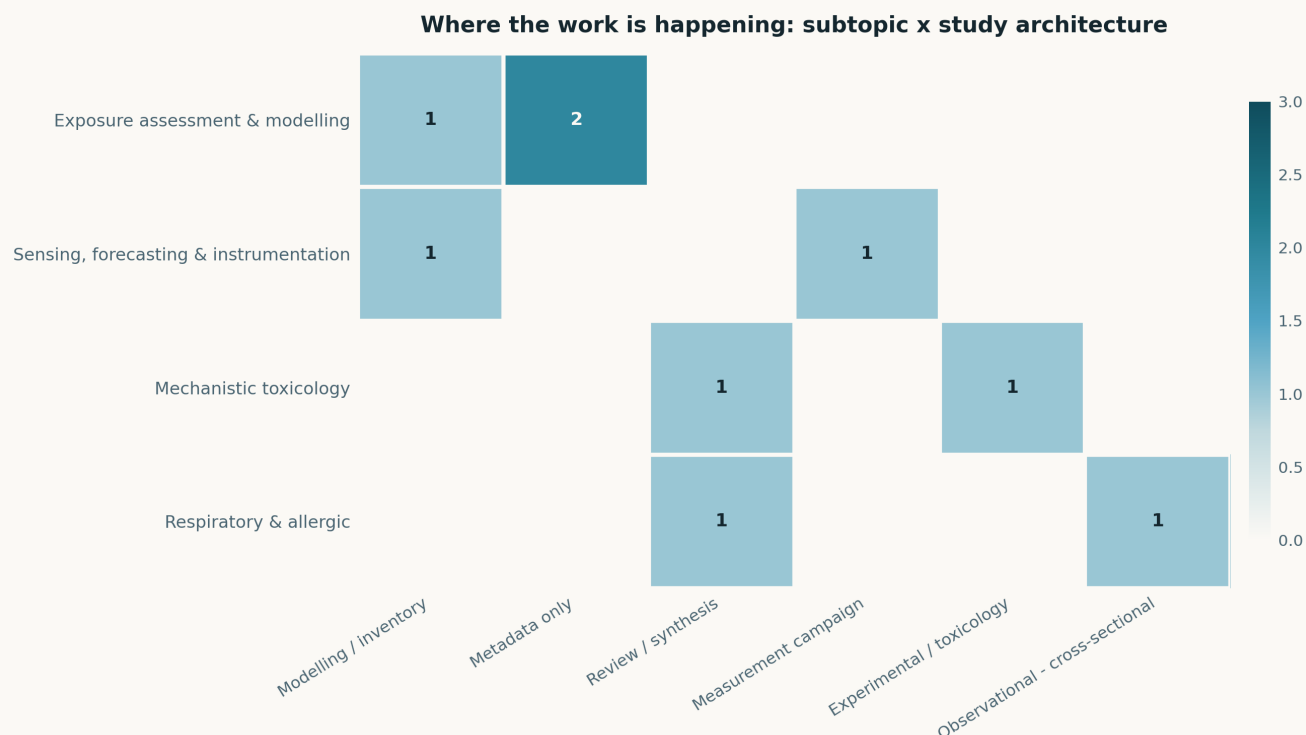


Fig. Subtopic $\times$ study-architecture cross-tabulation. The only cell holding more than one record is exposure assessment $\times$ Metadata only — i.e. the densest cell in the matrix is an artefact of two publisher deposits arriving without abstracts, not a methodological concentration. Every other occupied cell holds exactly one record, and the two narrative reviews sit in different rows (mechanistic toxicology and respiratory) despite arriving from the same journal issue. With nine records over six architectures the matrix is close to a permutation and reading a pattern into it would be over-interpretation.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

2 records

Neuberger et al. 2026 Atmos Chem Phys

Importance of hydrated aerosol particles for aerosol–fog relationships in the Italian Po Valley

FAIRARI 2021/2022 aerosol–fog observations at San Pietro Capofiume, analysed with κ -Köhler theory and the MIMICA large-eddy simulation. Median hygroscopicity separated by population: **fog residuals** $\kappa = 0.45$, **interstitial particles** **0.40**, **out-of-fog aerosol** **0.34**, the gradient reflecting inorganic enrichment in activated droplets. Excluding hydrated but non-activated particles from the analysis **inflated effective diameter by 81 %** ($11.6 \rightarrow 21.0 \mu\text{m}$) and **depressed droplet number concentration by 87 %** ($97.4 \rightarrow 12.4 \text{ cm}^{-3}$); hydrated particles supplied **21 % of liquid water content** and drove **36 % of sub-kilometre visibility events without any activation**. The LES leg shows fog prediction hinging on the largest dry particles. The limitation that binds is site specificity — one heavily polluted valley over two winters, with a κ distribution shaped by Po Valley nitrate that will not transfer to a marine or biomass-burning regime. For distributed low-cost sensing the transferable point is that the hydrated/activated distinction is a *composition* problem, so an RH correction fitted at one site during one season inherits that site's κ and no more.

doi: 10.5194/acp-26-11991-2026

Zhang et al. 2026c Atmos Chem Phys

Buffering of atmospheric nanoparticle growth by temperature-dependent shifts in molecular composition, volatility and diffusivity

A kinetic multiphase treatment applied to previously unreconciled field and chamber nanoparticle growth-rate data. The standing puzzle is that observed growth rates are smaller than theory predicts and *less* sensitive to condensable vapour concentration than theory predicts. The resolution offered is two antagonistic temperature dependencies: **slow surface-to-bulk transport limits vapour uptake by semi-solid, low-diffusivity particles**, while **Clausius–Clapeyron shifts in the volatility distribution enhance growth at low temperature and low concentration**. Netted out, they buffer the vapour-concentration dependence, which is why measured growth rates converge on a few nm/h across widely varying conditions. No new measurement is reported — this is a reinterpretation of existing datasets, and its persuasiveness rests on the volatility-distribution parameterisation rather than on independent validation. The relevance here is upstream of sensing: the 50–100 nm photochemical mode that Zhong et al. flagged yesterday as invisible to mass-based assessment is produced by exactly this growth process, and a mechanism that makes its rate insensitive to precursor loading also makes it insensitive to the emission controls that reduce $\text{PM}_{2.5}$ mass.

doi: 10.5194/acp-26-12037-2026

Exposure assessment & modelling

3 records

Zhang et al. 2026d Atmos Chem Phys*Improved representation of isoprene-derived secondary organic aerosol in CAM6-Chem reveals regional contrasts in its long-term changes over China*

An explicit isoprene-SOA formation scheme implemented in CAM6-Chem with high-NO_x gas-phase epoxide precursors and branch-resolved heterogeneous uptake from both low- and high-NO_x pathways. Against ground observations the update **improves the SOA normalised mean bias over China from −76.7 % to −51.6 %** and reproduces the compositional structure of four explicit ISOA subspecies. Over 2000–2019 the national mean ISOA trend is weak because regional changes cancel: **the sharpest increase is in Southwest China, where rising biogenic isoprene emission is the leading statistical predictor at a 58.3 % regression-based contribution**, with the strongest decrease in Shaanxi–Gansu–Ningxia. The honest reading of the headline number is that a −51.6 % bias is still a factor-of-two underestimate — the scheme narrows the gap without closing it, and a regression-based attribution is not a causal decomposition. For exposure work the consequence is directional: any health study using a CTM-derived organic-aerosol surface over China is working with a field that is known-low by roughly half, and non-uniformly so by region.

doi: 10.5194/acp-26-11967-2026

Kim et al. 2026 Atmos Environ*Application of machine learning-based wind corrections to coupled WRF–CMAQ simulations: a comparison of dynamic assimilation and static replacement*

Metadata only. Crossref by-ISSN sweep; title, authorship and DOI verified, no abstract deposited. The comparison it names — injecting an ML-corrected wind field into CMAQ dynamically during the run versus statically replacing the WRF field beforehand — is a live methodological question for anyone producing exposure surfaces, because wind error is the dominant transport error in urban CTM output and the two strategies have very different mass-conservation properties. Nothing is asserted about which won. Scholar Gateway cannot reach it (corpus ends May 2026) and Consensus has not indexed it. Second-priority abstract retry behind Werderman et al.

doi: 10.1016/j.j.atmosenv.2026.122324

Kang et al. 2026 Atmos Environ*Characteristics and dry deposition fluxes of particulate parent, alkyl, and nitro-polycyclic aromatic hydrocarbons in the Bohai Sea Strait, China*

Metadata only. Crossref sweep, no abstract deposited. Nitro-PAHs are among the more directly mutagenic particle-bound species and are under-measured relative to the parent compounds, and a marine-strait dry-deposition budget is a receptor-side measurement rather than an emission inventory — but no flux, no concentration and no sampling period are extractable from the deposited record. Flagged for retry, low priority relative to the wind-correction paper.

doi: 10.1016/j.j.atmosenv.2026.122322

Mechanistic toxicology**2 records****Li & Li 2026** Zhongguo Fei Ai Za Zhi*Research advances in the pulmonary carcinogenic effects of atmospheric fine particulate matter mediated by GPER*

Narrative review positioning the G protein-coupled estrogen receptor as the molecular hub linking PM_{2.5} to lung carcinogenesis. The argument: environmental estrogen-like organics and metal ions carried on PM_{2.5} activate GPER, which is overexpressed in lung tumour tissue, and the activated receptor drives **MAPK/ERK and PI3K/AKT** signalling into inflammation, oxidative stress and ferroptosis, with downstream proliferation, EMT and tumour-microenvironment remodelling. GPER inhibition is proposed as a target. **The weaknesses are structural to the format:** no search protocol, no risk-of-bias assessment, no quantitative synthesis, and a receptor-centric frame that cannot by construction weigh GPER against the AhR and oxidative-stress pathways it does not review. Read it as a hypothesis map, not as evidence of pathway dominance. The one genuinely actionable implication for exposure science is that it assigns the causal role to specific *components* — estrogenic organics and transition metals — rather than to mass, which is another argument for composition-resolved rather than gravimetric-equivalent monitoring.

doi: 10.3779/j.issn.1009-3419.2026.101.13

Qin et al. 2026 Biofactors*AlCl₃ aerosol exposure is associated with pulmonary lactate accumulation, neutrophil dysfunction, and fibrosis-like remodeling*

Repeated AlCl₃ aerosol exposure in rats, with and without the MCT1 inhibitor AZD3965, plus bone-marrow-derived neutrophils treated with sodium L-lactate, plus a supportive human cohort of aluminium-exposure-related ILD, non-aluminium ILD and healthy controls characterised by quantitative CT, spirometry and blood transcriptomics. Exposure produced pulmonary aluminium deposition, **lactate accumulation**, oxidative stress, DNA damage response signalling and collagen-rich fibrosis-like remodelling; in vitro, high lactate impaired neutrophil chemotaxis, oxidative burst, mitochondrial membrane potential, calcium signalling and NET responses. AZD3965 partially reversed both. Al-ILD patients showed greater CT parenchymal abnormality and lower FVC and DLCO than non-Al-ILD, with innate-immune, cytokine, ECM and NET pathways enriched in blood. **No exposure concentration, particle size distribution or duration appears in the deposited abstract**, so the animal arm cannot be placed on any dose axis, and the human arm is cross-sectional with no exposure measurement at all — it supports the mechanism narratively and cannot test it. Relevant here as occupational-aerosol toxicology and as a reminder that metal-bearing particulate has organ-level effects that mass concentration does not index.

doi: 10.1002/biof.70146

Respiratory & allergic**2 records****Zhongguo Fei Ai Za Zhi 2026** Zhongguo Fei Ai Za Zhi*China Wumeng Mountain lung cancer: evidence chain, clinical and molecular characteristics of regional lung cancer related to coal pollution in the Yunnan–Guizhou Plateau*

Narrative synthesis assembling an exposure → component → biological-effect → clinical-molecular → prevention chain for household smoky-coal lung cancer in Xuanwei/Fuyuan, searching PubMed, Web of Science, Embase, CNKI and Wanfang to May 2026. Carcinogenic components named: **PAHs and methylated PAHs, PM_{2.5}/black carbon, inorganic minerals and metals, and carbonyls**, as a mixture. The clinical phenotype is distinctive — predominantly **never-smoking women**, high adenocarcinoma fraction, young onset, early metastasis, multifocal disease and family aggregation. So is the molecular one: total EGFR mutation is high but classical 19del and L858R are *under-represented*, while **G719X, S768I, KRAS G12C and ALK/ROS1 fusions are elevated** relative to non-Wumeng lung cancer; ARHGEF5, MYO18B and MUC16 are proposed germline susceptibility genes interacting with POT1/TERT/TERF2. Sputum microbiome shifts are reported alongside. **It is a narrative review by an unnamed author group with no formal inclusion criteria or bias assessment**, drawing on the same group's prior work, so the evidence chain is an argument rather than a synthesis. Its value is as the clearest statement in this corpus of a testable source-composition-to-mutation-spectrum hypothesis.

doi: 10.3779/j.issn.1009-3419.2026.102.16

Aulia Anwar et al. 2026 *Exp Oncol**Smoking, environmental exposures, and chronic lung disease as key risk factors for lung cancer in North Sumatra*

Hospital-based case-control across three referral hospitals in North Sumatra, August–November 2024: **240 histologically confirmed lung cancers and 240 controls** without malignancy or chronic respiratory disease, matched on sex and residential area with age adjusted in the model. Independent predictors: **unhealthy indoor environment (poor ventilation, household pollutants) OR 2.53 (95% CI 1.45–4.40)** and **occupational exposure OR 1.91 (1.13–3.25)**; tuberculosis and COPD were more prevalent among cases. **Three weaknesses threaten the inference and should be stated first.** The exposure is a self-reported composite category with no measured particulate concentration, so it is subject to recall bias in a design where cases know their diagnosis. Controls were selected to be free of chronic respiratory disease, which is a collider on the pathway from household smoke to both COPD and lung cancer and biases the household-environment OR upward. And smoking differed between groups at $p = 0.023$ yet does not appear among the adjusted independent predictors — the deposited abstract does not explain that discrepancy. Treat 2.53 as an upper bound on a poorly specified exposure, not as a household-PM effect estimate.

doi: 10.15407/exp-oncology.2026.02.152

Corpus register

First author	Focus	Journal	Design	Metric	Region
Respiratory & allergic					
Aulia Anwar et al.	Smoking, environmental exposures, and chronic lung disease as key risk factors for lung cancer in North Sumatra	<i>Exp Oncol</i>	Case-control	Self-reported unhealthy indoor environment (ventilation, household pollutants)	Indonesia
Zhongguo Fei Ai Za Zhi	China Wumeng Mountain lung cancer: evidence chain, clinical and molecular characteristics of regional lung cancer related to coal pollution in the Yunnan-Guizhou Plateau	<i>Zhongguo Fei Ai Za Zhi</i>	Narrative review	Household smoky-coal PM _{2.5} / black carbon, PAHs and methylated PAHs	China
Mechanistic toxicology					
Li & Li	Research advances in the pulmonary carcinogenic effects of atmospheric fine particulate matter mediated by GPER	<i>Zhongguo Fei Ai Za Zhi</i>	Narrative review	PM _{2.5} estrogen-like organic components and metal ions	China
Qin et al.	AlCl ₃ aerosol exposure is associated with pulmonary lactate accumulation, neutrophil dysfunction, and fibrosis-like remodeling	<i>Biofactors</i>	Animal + in vitro + human cohort	Repeated AlCl ₃ aerosol inhalation (dose not deposited)	China
Sensing, forecasting & instrumentation					
Neuberger et al.	Importance of hydrated aerosol particles for aerosol-fog relationships in the Italian Po Valley	<i>Atmos Chem Phys</i>	Field campaign + LES modelling	Hygroscopicity parameter kappa; hydrated vs activated particle diameter	Italy
Zhang et al.	Buffering of atmospheric nanoparticle growth by temperature-dependent shifts in molecular composition, volatility and diffusivity	<i>Atmos Chem Phys</i>	Kinetic model + data reanalysis	Nanoparticle growth rate (nm/h) vs condensable organic vapour	Global / multi-region
Exposure assessment & modelling					
Kang et al.	Characteristics and dry deposition fluxes of particulate parent, alkyl, and nitro-polycyclic aromatic hydrocarbons in the Bohai Sea Strait, China	<i>Atmos Environ</i>	Metadata only (no abstract)	Particle-bound parent / alkyl / nitro-PAH (not extractable)	China
Kim et al.	Application of machine learning-based wind corrections to coupled WRF-CMAQ simulations: a comparison of dynamic assimilation and static replacement	<i>Atmos Environ</i>	Metadata only (no abstract)	Not extractable	Not stated
Zhang et al.	Improved representation of isoprene-derived secondary organic aerosol in CAM6-Chem reveals regional contrasts in its long-term changes over China	<i>Atmos Chem Phys</i>	Chemical transport modelling	Isoprene-derived SOA mass and subspecies composition	China

Provenance and method

Entry window **2026-08-24** → **2026-08-24**, contiguous with the 23 August issue. **This issue was built late**, on 25 August: the 24 August harvest ran and its cache was warmed, but the issue was deliberately deferred to the following scheduled run. The window is unchanged and no gap was left.

Sources. PubMed E-utilities on [EDAT] returned 3 records; the connector re-query on the health axis returned **4** and on the instrumentation axis **1**, five distinct records in total, of which four shipped. Europe PMC on CREATION_DATE returned **9 records, all archival PMC back-scans** with no title, DOI or abstract — the same structural artefact as the 13 seen on 23 August — and none shipped. The Crossref by-ISSN sweep returned 9 records and shipped 5, including the **first four *Atmospheric Chemistry and Physics* deposits the pipeline has ever caught**. OpenAlex returned an empty result for the 22nd consecutive run and remains a broken leg. arXiv returned nothing new under the recency screen. Consensus was swept on low-cost sensor calibration and drift and on satellite-surface PM_{2.5} fusion and returned 6 on-topic records, **all 6 already in seen.json**. Scholar Gateway was queried for the abstract-free WRF-CMAQ record and cannot reach it; its corpus ends May 2026. **Screening.** 21 records were screened out and written to `state/rejected.jsonl` with an individual reason each; the 9 archival back-scans carry a shared structural reason and the 6 Consensus hits carry the issue date on which each was originally shipped. One borderline exclusion is logged as `borderline`: a physics-informed indoor CO₂ prediction method whose architecture transfers directly to PM but which reports no particulate contrast. **Labels.** Two records carry no abstract and are marked *Metadata only*; their design, sample size, endpoint and study location are recorded as not extracted rather than inferred. **The forest plot carries two estimates from a single model** and both exposures are self-reported categories, stated on the axis; no pooled estimate is drawn. All DOIs were checked by `check_dois.py` before publication. Content derived from publisher metadata; abstracts remain the copyright of their respective publishers.